

Beetle Antennae Search guided RRT* for path planning of GIS inspection and maintenance robot *

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Abstract—Aiming at efficient path planning of the gas insulated switchgear (GIS) robot which is used for inspecting and maintaining the GIS equipment, this paper proposed a Beetle Antennae Search (BAS)-inspired RRT* algorithm. The new algorithm combines basic RRT* with BAS, a metaheuristic approach, to inspire the sampling, and utilizes a weighted lazy extension mechanism to contribute expanding useful nodes. BASL-RRT* is firstly compared with three improved RRT* algorithms and simulation results verify the effectiveness and the superiority of BASL-RRT*. Then, we apply this new method on path planning of GIS robot. The result shows that safe and smooth path could be obtained efficiently.

Keywords—GIS robot, inspection and maintenance, path planning, RRT*, beetle antennae search

I. INTRODUCTION

With the rapid growth of the power system, it has become a trend that closed GIS equipment with high degree of integration would replace those large open substations due to its high reliability and low failure rate [1]. But because of its own electrical and mechanical factors or human factors, a long time continuous work for the GIS equipment will cause the accumulation of foreign matters in the GIS cavity. Specifically, some powder-shape foreign matters would be generated by the GIS equipment internal flashover, insulators aging or the presence of conductive impurities, and in the installation or maintenance process of GIS equipment, some tool objects such as screws, nuts, and sand, metal chips would be left by mistake. It would certainly cause safety accident for the operation of GIS equipment and jeopardize the power production if the above-mentioned foreign bodies are not cleaned in time. Therefore, the cavity of GIS equipment must be cleaned and maintained, but it is tough for personnel to enter the cavity to finish the maintenance task, and the current common approach generally requires manual dismantling of the GIS equipment, which is complicated and inefficient. With the deep development of robot technology, related research on the automatic and intelligent operation robot suitable for the inspection and maintenance in GIS cavity become desirably possible. Fig. 1 shows an example of a GIS cavity.



Fig. 1. Example of a GIS cavity.

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Path planning is the key technology to achieve successful obstacle avoidance and control of maintenance robot in GIS cavity. The biggest difference of GIS robot is that the robot moves on the cylindrical cavity inner surface in GIS. The objective of path planning is to calculate the feasible path from the initial state to the target state, while ensuring the safety of avoiding collision with obstacles, and minimizing the cost function given such as length or other requirements [2]. A large number of path planning methods have been invented by scholars and successfully implemented in the actual robot operation, and have been modified and continuously improved to cope with special environments and tasks. These planning techniques, which are widely applied in the research and development of intelligent mobile agents such as AGV, USV, UAV and AUV in industrial occasions and social life [3-6], can be basically divided into two groups: classical methods and meta-heuristic methods [7].

Classical methods, which include Artificial Potential Field method, Graph-search-based methods and Sampling-based methods (SBP), are commonly used because their simplicity to implement. Graph-search-based schemes like A* and Dijkstra are castigated for their discrete transitions and exponentially increased computational requirements if the graph is discretized on a massive scale especially on GIS-like environment. Meta-heuristic methods especially those inspired by computational intelligence or biological swarm intelligence like PSO, DE, GA and ACO have also been demonstrated to become an effective path optimization and planning approach [7]. One of the large properties of the metaheuristic methods is the adoption of probabilistic decisions during iterations, which means they are always non-deterministic and the search may get stuck in local minimum so that some supplementary modified strategies should be usually embedded to avoid the drawback of local trap. In order to solve the issue of high dimensional planning, Sampling based planning methods (SBP), with a way of randomly sampling in configuration space to connect two ideal states, was invented, Sampling based methods successfully break through high dimensional limitation with the most famous initial two algorithms, Probabilistic Roadmap (PRM) and Rapidly-exploring Random Tree (RRT) [8-9]. Early, initial SBP methods couldn't meet the optimality of the solution path due to randomness. Considering this defect, RRT* was proposed, which adds two extra procedures called Choose Best Parent and Rewire to guarantee path optimality compared to origin RRT [10]. More variants of RRT* are quickly proposed to improve the Convergence speed and computational efficiency that two blocks of the scheme [11].

RRT*-Smart [12] and Informed-RRT* [13], both two variants limit the sampling zone after finding the first solution, where RRT*-Smart introduces the Beacons and Informed-RRT* the informed ellipsoidal subset heuristic.

Most of the robot applications of the above algorithms are concentrated in the two-dimensional and three-dimensional Euclidean space, and few applications are made to the path planning of maintenance robots moving on the cylindrical cavity inner surface in GIS.

In order to solve the maintenance robot path planning problem on GIS inner surface, we propose a new RRT* algorithm denoted as BASL-RRT* combined with the Beetle Antennae Search (BAS) [14], a novel meta-heuristic optimization, and introduce a lazy node extension mechanism to overcome disadvantages that RRT* possesses.

II. MODELLING OF THE ENVIRONMENT

A. The GIS cavity modelling

The internal wall of GIS cavity where the maintenance robot move is a curved and smooth cylindrical surface. despite the location of robot will change in three dimensions, as a result of the limitation of degrees of freedom of robot itself, the robot should stick on and never get out of the surface of cavity wall in GIS, based on this, we do path planning by unfolding GIS cavity surface into a two-dimensional plane and set up the new corresponding coordinate system, hence, only two coordinates will only be considered in the path planning procedure. The transformation is presented in Fig. 2, where the relationship between two coordinate systems, the GIS cavity surface and the new 2D plane, is illustrated.

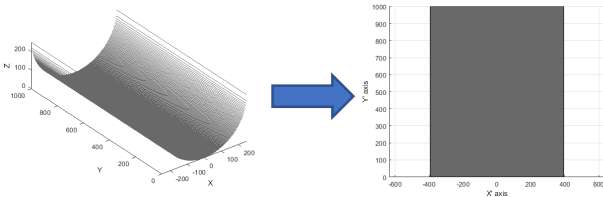


Fig. 2. Unfolding of a surface of a GIS cavity.

B. Obstacles modelling

When the maintenance robot performs tasks in the GIS cavity, it needs to avoid the dangerous areas, and the dangerous areas inside the GIS cavity are mainly the round hole in the inner wall and other foreign bodies remaining on the surface of the cavity. The round hole and all foreign bodies are regarded as obstacles for robot, all obstacles are modelled as circles in 2D plane. On the basis of the environment transformation and obstacle model, path planning of the maintenance robot is to compute a feasible optimal path with a minimum cost that connects the initial and goal states while keeping a safe motion from obstacles.

III. RELATED ALGORITHMS

A. RRT*

RRT* is a sort of SBP method, that means this algorithm will randomly sample in configuration space to construct a tree, different from original RRT, RRT* will not stop sampling if the tree successfully reach the goal region but to continue this sample process, in which choose parent and rewire, these two distinct steps of RRT*, will keep modifying

and optimizing the whole tree structure to guarantee asymptotic optimality.

Let X be the configuration space in which robot operates, the region safe for robot navigation in which robot is able to move is denoted as X_{free} , and other remaining region in the configuration space is X_{obs} where no edges of the tree could pass the collision check, finally, x_{init} and x_{goal} are defined to stand for the starting and end state. RRT* algorithm starts running by inserting x_{init} , as a root node, into the empty tree. Then, in each next iteration to come, RRT* operates a sample process by sampling a random node called x_{rand} within X_{free} , and RRT* will search the whole tree to spot a nearest node, $x_{nearest}$, and generate a new node, x_{new} , which is just a step size long from $x_{nearest}$ in the direction of x_{rand} , and if the segment between $x_{nearest}$ and x_{new} pass the collision check, the two states will be connected and x_{new} will be added into the tree at instantly because they are safe. So far, abovementioned procedure is same as expansion process of basic RRT, however, RRT* introduces two further extra processes, the one is to make an attempt to update the parent node of newly added node x_{new} and the other is to view whether it is possible to update respectively those parents of nodes within a specific circle centered at x_{new} , from their origin parents to x_{new} . Explicitly, first, RRT* needs to identify a set of neighborhood nodes those within a circle centered at x_{new} , with a special radius. The set of neighborhood nodes is denoted as X_{near} and all members in this set will be confirmed whether it could be a better-cost parent node, if a satisfied node is caught, this node will be denoted as x_{min} , and $x_{nearest}$, the parent node of x_{new} , will be replaced by the node x_{min} so that old connection has been updated. Second, another extra process Rewire will be executed, all neighborhood nodes in X_{near} will try to apply the newly added node x_{new} as their respective parent node instead of old parent node as soon as possible, such a replacement will happen on the condition that a lower cost would be attained after using x_{new} as a new parent node.

B. Beetle antenna search

Beetle antenna search (BAS) is a novel metaheuristic optimization algorithm inspired by the foraging behavior of beetles. It has a simple structure but a great search speed in finding optimal solution. Its effective performance render it an extensive application in many fields such as PID turning, machine learning and path planning, etc. [15].

The basic principle of BAS is that the optimization process will be treated as a beetle foraging behavior, the beetle will judge the direction of their next movement based on the concentration of food smell perceived by their own antennae. At beginning, the beetle is unaware of its final search destination, and only greedily looks to the tentacle that has picked up a stronger scent of food to determine its next move. The concentration of food is abstracted as a value of the fitness function, the stochastic search direction of the beetle can be expressed by the formula:

$$d = \frac{rand(k,1)}{\|rand(k,1)\|} \quad (1)$$

Where k is the number of spatial dimensions, once the search direction has been determined, the left and right antennae of the beetle will detect in the corresponding positions.

$$\begin{cases} x_r = x' + \delta' d \\ x_l = x' - \delta' d \end{cases} \quad (2)$$

Where x_l and x_r are current positions of the left and right antenna, x' is the current position of the center of mass of the beetle, δ' is the length of every antenna which influences the search range. The beetle will go on searching in the direction where the fitness is lower, this behavior can be modeled by the flowing formula:

$$x' = x'^{-1} + step' \cdot d \cdot \text{sign}(f(x_r) - f(x_l)) \quad (3)$$

Where $step'$ is the step size of beetle movement in current iteration, $\text{sign}(\)$ is the sign function, $f(x_r)$ and $f(x_l)$ are fitness values of the two beetle's antennae. Then, the beetle will reach a new position, so far the search of one iteration has been completed, the initialized length of antenna and step size should be long enough so that a larger unknown environment will be explored, however, the length will gradually decrease with time t according following formula:

$$\begin{aligned} \delta' &= 0.95\delta'^{-1} + 0.01 \\ step' &= 0.95step'^{-1} \end{aligned} \quad (4)$$

The above decreasing approach is to achieve as much trade off as possible between global exploration and local exploitation.

IV. PROPOSED METHOD

RRT* is a **breakthrough of modification** for basic RRT because of its path optimality, however, this algorithm has one **overwhelming defect** that it **keep a totally random search in the whole configuration space which is a hinder of finding the initial solution** and converging to final optimal solution or asymptotically optimal solution. In consideration of that weakness, we present a **hybrid method called BASL-RRT* associated original RRT* with BAS and introduce a weighted lazy extension mechanism to accelerate RRT***.

A. BAS guidance

Goal-biasing strategy is always applied to guide the growth of the RRT tree [16], it makes tree extend to the goal point directly in some iterations with a specific probability, it is effective in most dense cases, but it will get stuck in local minimum in some non-convex occasions. In this paper, we utilize BAS to drive RRT tree to grow toward to a vicinity around the goal region. A more detailed explanation is as follows:

We assume that the target point is where the food exists, and the beetle will stop at the target point after successful search. Here, we stipulate that when the beetle finally stops, the connection between the ends of the two tentacles is perpendicular to the connection between the starting point and the target point. The ideal region is defined as the vicinity of the two tentacles, and the beetle starts up from the initial point and flies towards these two points, the objective function optimized by BAS is defined as:

$$fit = \|x^t - x_{goalL}\| + \|x^t - x_{goalR}\| \quad (5)$$

Where, x_{goalL} and x_{goalR} are the positions of the two tentacles of the beetle on the left and right sides of the target point respectively. Our hybrid approach is to call the BAS algorithm to obtain the bias to guide the extension of the tree in each iteration before RRT* finds the initial feasible path. Since BAS will get the optimal solution as the number of iterations of BAS algorithm increases, which indicate that the biasing point will be approximated to the line segment connected by x_{goalL} and x_{goalR} . In order to realize that the sampling biasing points are concentrated near the line and more diverse sampling states can be obtained to avoid local minima, we controlled the number of iterations in the BAS algorithm to prevent BAS from obtaining the optimal solution, that is, to reduce the number of iterations of BAS so that the solution of BAS could be attracted near the line instead of reaching the minimum position.

B. Weighted lazy extension

RRT* will continuously grow new branches and reconnect local edges of whole tree to achieve optimality, but some of the new edges can't help improve the path due to its global uniform sampling, for the sake of enhancing quality of newly added nodes and edges, in this paper, we proposed a weighted lazy extension mechanism, we introduce an extra process before inserting the node, that is an evaluation of the performance of the potential new node. Here comes a full explanation, when we acquire the node and the direction from it to node, the mechanism will calculate a sum of the Cost to Come of $x_{nearest}$ and the step size with their weights, the evaluation can be described by

$$Cost_{evl} = w_1 c_1 \cdot Cost_{toCome} + w_2 c_2 \cdot stepsize \quad (6)$$

If $Cost_{evl}$ is larger than current path cost best, the next extension process will be ceased, which means this iteration will terminate immediately while three steps, collision check, choose best parent and rewire, will not be executed so it save the computation time of all iterations.

C. BASL-RRT*

The details of our proposed BASL-RRT* is described in Algorithm 1, BASL-RRT* has two main characteristics different from basic RRT* in that it generates sample node by BAS algorithm with a probability (Line 5-6 in Alg.1) to avoid local minima caused by goal biasing, and the proposed weighted lazy extension mechanism (Line 12-13 in Alg.1) will be activated once the initial solution is found, which can eliminate inferior new edges those can't contribute to the path optimization.

Algorithm 1: BASL-RRT*

1. $T \leftarrow \text{EmptyTree}(\)$
 2. $V \leftarrow \{x_{init}\}; E \leftarrow \emptyset; T \leftarrow (V, E);$
 3. $PathLength \leftarrow \infty$
 4. **for** $i = 0$ to $i = N$ **do**
 5. **if** $PathLength = \infty \cap RAND < P$ **then**
 6. $x_{rand} \leftarrow \text{BeetleAntennaeSearchSample}(i)$
 7. **else**
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8.    $x_{rand} \leftarrow \text{UniformSample}(i);$ 
9.   end if
10.   $x_{nearest} \leftarrow \text{NearestVertex}(T, x_{rand});$ 
11.   $x_{new} \leftarrow \text{Steer}(x_{nearest}, x_{rand});$ 
12.   $\text{cost}_{elv}(x_{new}) = \omega_1 \text{cost}_{locom} + \omega_2 c_2 \text{step}$ 
13.  if  $\text{cost}_{elv}(x_{new}) < \text{PathLength}$  then
14.    if  $\text{ObstacleFree}(x_{nearest}, x_{new})$  then
15.       $X_{near} \leftarrow \text{NearVertices}(T, x_{new}, r);$ 
16.       $x_{min} \leftarrow \text{ChooseBestParent}(X_{near}, x_{nearest}, x_{new});$ 
17.       $V \leftarrow V \cup \{x_{min}\};$ 
18.       $E \leftarrow E \cup \{x_{nearest}, x_{min}\};$ 
19.       $T \leftarrow \text{Rewire}(T, X_{near}, x_{min}, x_{new})$ 
20.    end if
21.  end if
22. end for
23. return  $T$ 

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D. Post pruning and smooth technology

Path generated by RRT* usually has some sharp angles which are not smooth for robots. In order to meet the requirements of the turning of GIS maintain robot smoothly, we use B-spline curve to modify the path, its form is defined as

$$p(u) = \sum_{i=0}^n P_i \cdot N_{i,k}(u) \quad (7)$$

Where $u \in [0,1]$ is the knots vector which has $n+k+2$ elements, $N_{i,k}(u)$ are the basis functions of degree k , and P_i are control points caught by RRT* result. The details of our proposed BASL-RRT* is described in Algorithm 1, BASL-RRT* has two main characteristics different from

basic RRT* in that it generates sample node by BAS algorithm with a probability to avoid local minima caused by goal biasing (Line 5-6 in Alg.1), and the proposed weighted lazy extension mechanism will be activated once the initial solution is found (Line 12-13 in Alg.1), which can eliminate inferior new edges those can't contribute to the path optimization.

V. SIMULATION EXPERIMENTS AND ANALYSIS

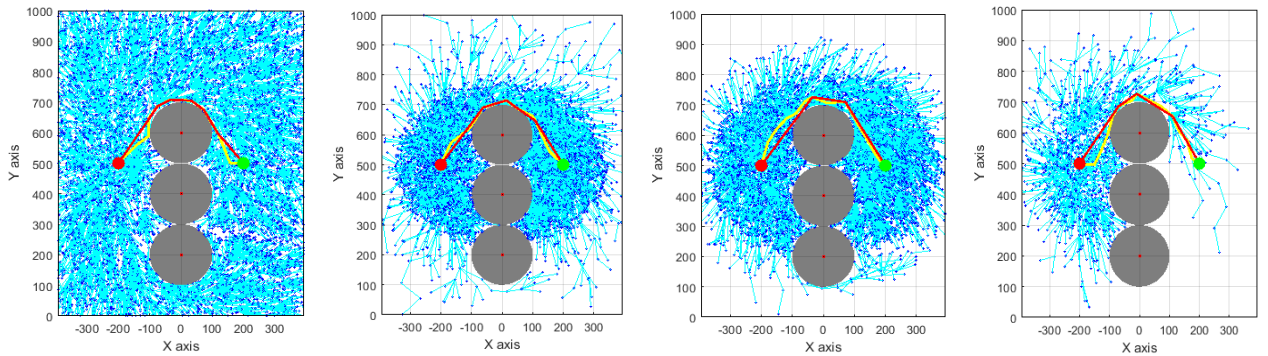
In this section, our proposed algorithm is simulated in matlab2018b, and the parameters used for our proposed are as follows:

$$w_1 = 0.5, w_2 = 0.5, c_1 = 2, c_2 = \frac{\|x_{new} - x_{goal}\|}{0.5 \text{stepsize}} + 2 \quad (8)$$

we at first test BAS-RRT* in 2 kinds of 2D maps, simple map and complex map, which verifies the feasibility and superiority of BAS-RRT*, meanwhile, the proposed method is compared with basic RRT*, informed-RRT* and Gb-informed-RRT* which is a representative well-performed version of RRT* variants. This test removed post smooth stage to ensure the fairness with other two algorithms. The maps are shown in Fig. 3 and Fig. 4, the solution acquired by three algorithms is provided as well, the performance statistics of the four in the two maps is recorded in Table 1, where initial cost and initial time are cost and time respectively when the algorithm finds its initial path. All experiments are executed 10 times on each map and the average is recorded.

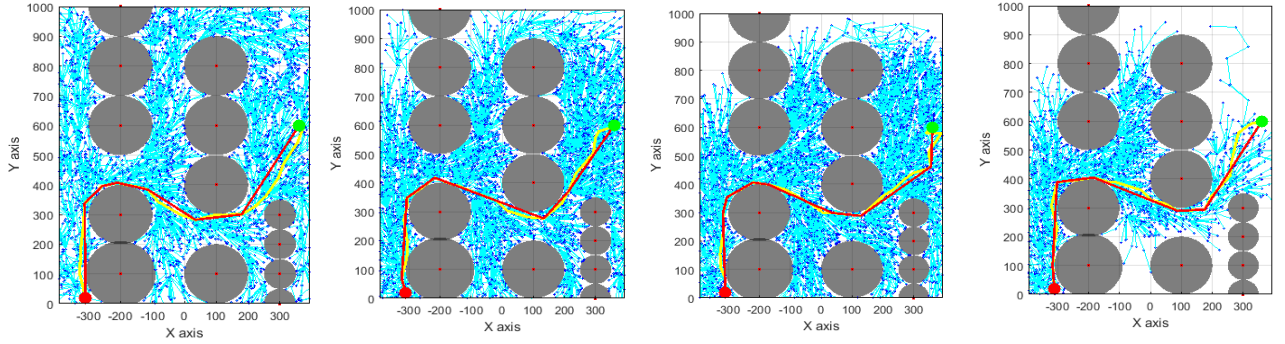
A. Analysis of the two enrionments tests

It can be seen from Fig. 3, Fig. 4 and Table 1 that our proposed BASL-RRT* acquires better solution in both two maps. It is worth noting that BASL-RRT* overcomes the blindness in basic RRT* by merging BAS approach meanwhile the BAS activated biasing is better than Goal-biasing approach in Gb-Informed-RRT*. Moreover, BASL-RRT* explores even less space than Informed-RRT* and finally much less nodes can be added thanks to the weighted lazy extension mechanism.



(a) RRT* in Map1 (b) Informed-RRT* in Map1 (c) Gb-Informed-RRT* in Map1 (d) BASL-RRT* in Map1

Fig. 3. The paths generated by the four algorithms in Map1.



(a) RRT* in Map2 (b) Informed-RRT* in Map2 (c) Gb-Informed-RRT* in Map2 (d) BASL-RRT* in Map2

Fig. 4. The paths generated by the four algorithms in Map2.

TABLE I. THE RESULTS OF THE FOUR ALGORITHMS

Map	Algorithm	Final cost/Initial cost	Running time/Initial time	Nodes	Std of Final cost
Map 1	RRT*	661.01/1154.84	133.15/2.28	8487	21.58
	Informed-RRT*	630.83/1103.14	69.07/1.75	6841	20.23
	Gb-Informed	629.83/962.93	66.93/1.68	6780	12.65
	BASL-RRT*	611.83/1015.82	29.53/1.43	1302	9.81
Map 2	RRT*	1276.79/1634.18	69.80/2.81	5161	26.34
	Informed-RRT*	1262.41/1654.08	61.83/2.62	4904	21.59
	Gb-Informed	1262.43/1633.09	69.45/0.83	4918	15.41
	BASL-RRT*	1231.88/1637.92	22.18/0.43	1962	17.57

B. GIS cavity simulation

Then our BAS-RRT* is applied in a GIS cavity occasion when the maintenance robot operate own task and all obstacles and the center hole are expanded according to the maximum radius of the robot so the robot can be treated as a mass point, the environment of GIS cavity and the smooth path is shown in Fig. 5. In this case, the path generated by BASL-RRT* is executed with the post path pruning and smoothed by B-spline curve to guarantee the smoothness of the real robot.

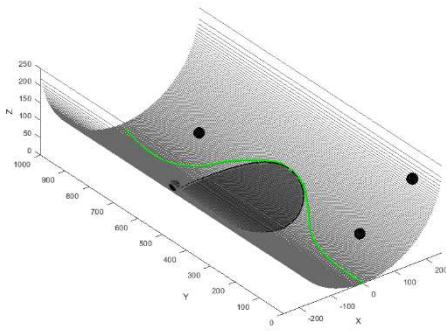


Fig. 5. The path generated by BASL-RRT* in GIS cavity.

VI. CONCLUSION

This paper presents a new method called BASL-RRT* for GIS maintenance robot path planning to deal with obstacle avoidance in cleaning task, based on BAS, more diverse biasing points can be used to guide the tree, and a weighted lazy extension of the newly added nodes help the algorithm identify potential superior nodes, and the smoothness have been considered to comfort the robot with B-spline curve. Besides, this paper has verified the advantages of BASL-

RRT* compared with other RRT* algorithms, moreover, BAS-RRT* has been applied in the GIS cavity case simulation, this is a relatively recent research on path planning of GIS maintenance robot that can successfully get a safe and effective path by the proposed BAS-hybridized approach.

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